

# Machine Learning–Driven Bitcoin Price Forecasting System

## Project Proposal

### Submitted by

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### 1. Introduction and Background

Due to their high volatility, cryptocurrencies such as Bitcoin have attracted much attention from investors due to the potential profit they bring. Traditional methods for forecasting future monetary developments are challenged by the complex, non-linear, and highly dynamic nature of cryptocurrency price movements.

The purpose of this project is to design an effective Machine Learning-based solution for predicting the price of Bitcoin using historical information and powerful forecasting techniques. Data-driven approaches are going to be applied to discover trends, patterns, and relationships within the time series.

This algorithm would be practical in real-world applications such as algorithmic trading, investment analysis, and fintech. Price prediction accuracy would allow investors to make well-informed decisions, minimize risk, and create more profitable trade strategies.

## **2. Problem Description and Project Objective**

Bitcoin price prediction is challenging due to:

- Market changes that happen quickly and are very volatile
- The effect of things outside of the business, like news, rules, and how investors feel
- Behavior of financial data that is not linear and changes over time

Traditional models don't adapt quickly enough to these kinds of problems, which makes their predictions wrong.

Also, making an automated forecasting system comes with problems like:

- Handling noisy and incomplete data
- Preventing overfitting in complex models
- Ensuring scalability and capability to predict in real-time

We want to create and put into use a machine learning-based system that can accurately and quickly predict Bitcoin prices using historical time-series data.

## **3. System Architecture**

The proposed Bitcoin price prediction system follows a modular pipeline that directly aligns with the project's implementation modules and the flow shown in the architecture diagram.

High-Level Architecture

The system is structured into three main stages: DATA, MODELING, and EVALUATION.

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### ***DATA Stage***

- Historical Bitcoin price data (Date, Open, High, Low, Close, Volume) is collected from public sources such as Investing.com or Kaggle and loaded as a CSV file into the project folder.

The data then passes through:

- Data Preprocessing (cleaning missing values and aggregating into daily records)
- Feature Engineering & Selection (including technical indicators: EMA, RSI, MACD, MOM, PROC, and %K)
- Rolling Window Transform
- Train-Test Split

### ***MODELING Stage***

- This stage includes:
  - Model Selection (Linear Regression, RBF SVM, Random Forest, kNN, Decision Tree, Elastic Net, and LSTM)
  - Training
  - Hyperparameter Tuning
- Models are trained for both price regression and directional classification (up/down).

### ***EVALUATION Stage***

- Model performance is assessed through:
  - Backtesting
  - Walk-Forward Validation
  - Performance Metrics (RMSE, MAE,  $R^2$  for regression; Accuracy, Precision, Recall, and confusion matrix for classification)

### ***User Interface***

A simulation dashboard will be developed using a React frontend connected to a lightweight Flask backend. The dashboard will allow users to visualize historical trends, compare model results, view actual vs. predicted prices, and interact with prediction outputs.

### **Flow:**



## 4. Methodology

The proposed Bitcoin price prediction system will be developed following a structured machine learning pipeline consisting of prediction techniques, mathematical modeling, data handling, feature engineering, model development, evaluation, and deployment.

### 4.1 Prediction Techniques

The following machine learning approaches will be implemented:

#### A. Linear Regression Model

Linear regression will be used as a linear approach to model the relationship between the dependent variable (Bitcoin closing price) and one or more independent variables. Both simple and multiple linear regression will be applied depending on the number of predictors.

#### B. RBF SVM

Support Vector Machines with Radial Basis Function (RBF) kernel will be employed. The RBF kernel nonlinearly maps samples into a higher-dimensional space, enabling it to handle nonlinear relationships between class labels and attributes. RBF is chosen as the primary kernel due to its reasonable first-choice performance, fewer hyperparameters compared to the polynomial kernel, and fewer numerical difficulties ( $0 < K_{ij} \leq 1$ ). Linear and polynomial kernels will also be tested for comparison.

#### C. Random Forests

Random Forest, an ensemble learning method, will be implemented by combining multiple decision trees. It fits various decision tree classifiers on different sub-samples of the dataset and uses averaging to improve predictive accuracy and control overfitting. Specifically, three time-series datasets will be constructed for 30, 60, and 120 minutes (corresponding to 180, 360, and 720 data points) prior to the current point. Separate models will be trained on these datasets to predict price changes ( $\Delta P1$  and  $\Delta P2$ ).

#### D. k-Nearest Neighbors (kNN)

The kNN algorithm will be used for both regression and classification tasks. It stores all available cases and assigns new instances to the class most common among its k nearest neighbors, using distance metrics such as Euclidean, Manhattan, Minkowski, or Hamming distance.

#### **E. Decision Tree**

Decision Tree, a supervised learning algorithm, will be utilized for both classification and regression. The algorithm splits the population into homogeneous sets based on the most significant independent variables to create distinct groups.

#### **F. Elastic Net**

Elastic Net regression, which combines the properties of LASSO and Ridge regression, will be applied to handle multicollinearity among predictors. It will regularize and select the most important variables to build a simpler and more accurate predictive model, especially suitable for non-stationary and non-linear time-series data.

#### **G. LSTM Model**

Long Short-Term Memory (LSTM) networks will be implemented to capture long-term dependencies in the time-series data.

#### **Mathematical Model**

Let  $(x^{(i)}, y^{(i)})$  represent a single observation. The set of samples is defined

as:  $S = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

The feature vector at time t is constructed

These feature vectors are stacked into matrix  $X \in \mathbb{R}^{n \times m}$ , and the target matrix is  $Y = [y^{(1)}, y^{(2)}, \dots, y^{(m)}]^T$ .

### **4.2 Features**

- Historical Data: Closing price (C) and trading Volume (V).
- Technical Indicators:
  - Exponential Moving Average (EMA)
  - Moving Average Convergence Divergence (MACD)
  - Relative Strength Index (RSI) at different periods
  - Momentum (MOM)
  - Price Rate of Change (PROC)
  - Stochastic Oscillator (%K)

These features will be computed over multiple time windows to capture trend, momentum, and volatility patterns.

### **4.3 Implementation Modules**

**The project will follow these sequential modules (tentative):**

1. **Data Collection** Historical Bitcoin price data will be collected, including Date, Open, High, Low, Close, and Volume. The dataset is expected to contain over 4,000 records spanning multiple years.
2. **Data Pre-processing** Missing values will be removed, and minute-level data (if collected) will be aggregated into daily records using group-by operations. All subsequent modeling will be performed on the cleaned daily dataset.
3. **Feature Selection and Engineering** Correlation analysis will be conducted among variables. Technical indicators (EMA, RSI, MACD, MOM, PROC, %K) will be engineered and added to the feature set. Feature importance will be evaluated using Random Forest.
4. **Problem Formulation** A log difference variable ( $\log\_diff$ ) will be created as  $\log(Close_t) - \log(Close_{t-1})$ . A binary directional target ( $Direction\_up$ ) will be defined: 1 if  $\log\_diff > 0$ , otherwise 0. Models will be trained for both price regression and direction classification.
5. **Training and Test Set Split** Different train-test ratios will be used: 80:20 for traditional models (Linear Regression, SVM, Random Forest) and 50:50 for neural network models (LSTM).
6. **Model Training** All prediction techniques (Linear Regression, RBF SVM, Random Forest, kNN, Decision Tree, Elastic Net, and LSTM) will be trained and compared. Hyperparameters for LSTM will include time step = 10, ReLU activation, 100 epochs, MSE loss, and batch size = 32.
7. **Evaluation** Models will be evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE),  $R^2$  for regression, and Accuracy, Precision, Recall, and confusion matrix for classification. Cross-validation will be performed to assess generalization. Comparative tables and actual vs. predicted price plots will be generated.
8. **Prediction and Deployment** The best-performing model will be used to generate price and direction predictions. A user-friendly simulation dashboard will be developed using Flask to compare different models, visualize results, and support decision-making for traders and researchers.

## 5. Scope

### In-Scope:

- Data collection, pre-processing, feature engineering, and technical indicator calculation
- Implementation and comparison of all listed prediction techniques
- Model training, evaluation, and result visualization
- Development of a simulation dashboard for model comparison

### Out-of-Scope:

- Real-time trading execution or live financial transactions
- Integration with external news or sentiment analysis (can be considered as future work)

## AI Ethics Considerations

- Only publicly available historical datasets will be used.
- Model predictions will be presented with appropriate uncertainty measures and will not be treated as financial advice.

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- Transparency will be maintained through feature importance analysis, evaluation metrics, and reproducible code.

## 6. Tools and Technology

- Programming Language: Python 3.x
- Libraries: Scikit-learn, TensorFlow/Keras, Pandas, NumPy, Matplotlib, Seaborn
- Development Environment: Jupyter Notebook / VS Code
- Deployment: Flask (for the simulation dashboard); FastAPI/Streamlit as optional enhancement

This methodology provides a comprehensive and robust framework for developing an effective Bitcoin price prediction system using advanced machine learning techniques.

## 7. Data Sources:

<https://www.kaggle.com/datasets/mczielinski/bitcoin-historical-data>

## 8. Hardware

- Minimum 8GB RAM
- GPU (optional for deep learning)

## 9. Contributions

| Name             | Contribution                                                                                                                                                                                            |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Esha Saeed       | <b>Data Engineer:</b> In charge of gathering, cleaning, preprocessing, and exploratory data analysis (EDA). Also works on feature engineering, which includes technical indicators like RSI and MACD.   |
| Muzammil Mahmood | <b>ML Engineer:</b> In charge of putting into action baseline models (Linear Regression, ARIMA) and advanced models (Random Forest, XGBoost, LSTM), as well as training, tuning, and evaluating models. |
| Khadija Masood   | <b>System Developer:</b> In charge of making the API (FastAPI), integrating the model into a system, and making a visualization dashboard (Streamlit) for users to interact with.                       |

## References

- [1] Chen, J. (2023). Analysis of Bitcoin Price Prediction Using Machine Learning. *Journal of Risk and Financial Management*, 16(1), 51.
- [2] Mallqui, D. C., & Fernandes, R. A. (2020). Bitcoin price prediction using machine learning: An approach to sample dimension engineering. *Journal of Computational and Applied Mathematics*, 365, 112395.
- [3] Ibrahim, S. K., & Singh, P. (2022). Bitcoin Price Prediction Using Machine Learning Techniques. *Journal of Informatics Electrical and Electronics Engineering*.
- [4] Gurupradeep, G., et al. (2023). Cryptocurrency Price Prediction using Machine Learning. *International Journal of Advanced Research in Computer and Communication Engineering*.
- [5] Kaur, G., et al. (2024). Predictive Modeling of Bitcoin Prices using Machine Learning Techniques. *International Journal of Intelligent Systems and Applications in Engineering*.



